

1. If $f(x) = \sqrt{x+5} - \sin\left(\frac{\pi}{2}x\right)$, then find the exact value of $f\left(\frac{2}{3}\right)$

2. Find the domain of each function:

(a) $f(x) = \sqrt{2-5x}$

(b) $f(x) = \frac{1}{x\sqrt{x+3}}$

(c) $f(x) = \sqrt[3]{2-7x}$

3. Find the Difference Quotient of each function:

(a) $f(x) = 2x^2 + x$

(b) $f(x) = \frac{1}{x+1}$
